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**Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology**  
**School of Computing**

**Department of Computer Science and Engineering**

**10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)**

**END SEMESTER PROJECT REVIEW**

**Date: 06-05-2026**

**JOB PORTAL MANAGEMENT SYSTEM**

**SDG: SDG No – SDG Title(Example: SDG 4 – Quality Education)**

**Presented By:**

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**Courses Handling Faculty:**

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# INTRODUCTION

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- Job Portal Management System is a web-based platform that connects employers and job seekers (students/users) in a centralized environment.
- It simplifies the recruitment process by allowing employers to post job vacancies and manage applications efficiently.
- Job seekers can browse available jobs, upload or update resumes, and apply for suitable positions
- The system provides features like shortlisting candidates, making hiring faster and more organized.
- It reduces the need for manual recruitment processes and improves communication between employers and applicants.
- The platform ensures easy access to job opportunities anytime and anywhere.
- It enhances transparency, reduces hiring time, and improves overall efficiency in the recruitment process.
- This system supports employment growth and aligns with SDG 8: Decent Work and Economic Growth by promoting job accessibility.

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# PROBLEM STATEMENT

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In today's competitive job market, students and job seekers often struggle to find relevant job opportunities in a timely manner, while employers face difficulty in identifying suitable candidates efficiently. The existing recruitment processes are often fragmented, time-consuming, and lack a centralized platform for seamless interaction between job providers and applicants.

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To address these challenges, a Job Portal Management System is required to provide an integrated solution where employers can host job openings and shortlist candidates, and students/job seekers can browse jobs, upload or update resumes, and apply for suitable positions. The system aims to streamline the recruitment process by enabling easy communication, efficient candidate management, and quick access to opportunities.



# OBJECTIVE(S)

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- **Centralized Web Platform:** A single, unified hub connecting job seekers and employers directly.
- **Role-Based Access Control:** Dedicated, secure dashboards for Students, Employers, and Admins.
- **Modern & Responsive UI:** Built with React.js using an engaging, high-contrast Neo-Brutalist design.
- **Automated Tracking:** Replaces messy email applications with real- time status updates (Pending, Shortlisted, Rejected).
- **High Security:** Implements Spring Security & JWT for stateless, secure authentication and data protection.
- **Scalable Architecture:** RESTful APIs using Spring Boot and MySQL allow for fast data retrieval and future expansion.
- **Streamlined Hiring Workflow:** Enables digital resume uploads, advanced job filtering, and instant applicant reviews.



# PROPOSED SYSTEM

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- **Full-Stack Architecture** : A web application using Spring Boot and React to connect employers and job seekers efficiently.

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- **Unified Recruitment Workflow** : Combines job posting, applicant tracking, shortlisting, and resume management in one system.

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- **AI-Powered Employer Tools** : Provides an AI-based question generator for creating role-specific technical assessments.
- **Eliminates Hiring Fragmentation** : Centralizes recruitment processes to reduce scattered workflows and improve organization.
- **Resolves Communication Delays** : Offers real-time updates and automated notifications for better user communication.
- **Superior Modern UI/UX** : Uses a responsive Neo-Brutalist design for a clean and engaging user experience. **Distinct Role-Based Dashboards** : Provides separate dashboards for students and employers for better usability. **Secure and Scalable Ecosystem** : Uses JWT authentication and admin controls to ensure security and scalability.

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# PROPOSED SYSTEM ARCHITECTURE

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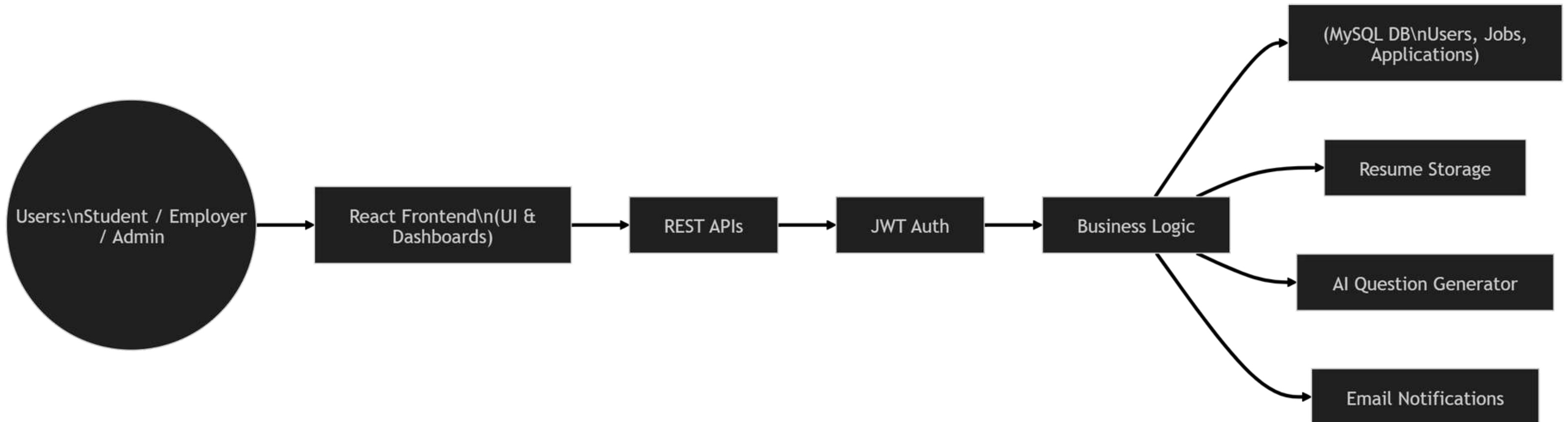
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# WORKFLOW

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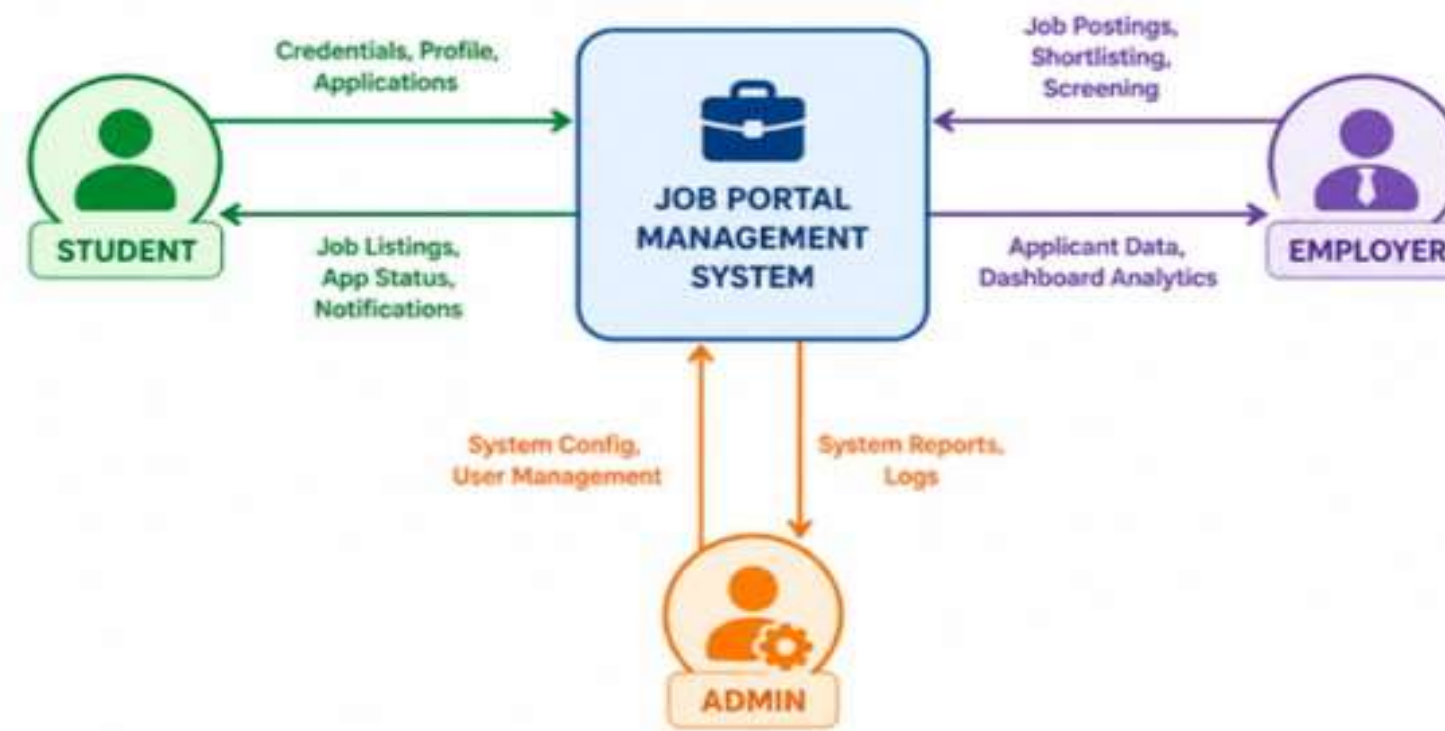
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# TECHNOLOGIES USED

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- **Frontend** : Built using React.js with Vite for fast development and optimized performance. React Router enables seamless role-based navigation, while Tailwind/Vanilla CSS provides a modern Neo- Brutalist UI. Axios is used for efficient communication with backend APIs.
- **Backend** : Developed using Java and Spring Boot to create scalable and secure REST APIs. Spring Security handles JWT-based authentication and role authorization. JPA/Hibernate simplifies database operations and ensures efficient data persistence.
- **Database** : MySQL is used as a relational database to store user details, job postings, and application data. It ensures data consistency, structured storage, and efficient querying. The system supports reliable data management for recruitment workflows.
- **Third-Party APIs** : Google Gemini API is integrated to generate dynamic, role-specific interview questions. SMTP services using Gmail are used for sending automated emails and OTP verification. This improves communication and user engagement.
- **Tools** : Development is done using Visual Studio Code and IntelliJ IDEA for coding and debugging. Git and GitHub manage version control. Postman and Swagger UI are used for API testing and documentation.
- **Package Managers** : npm is used to manage frontend dependencies and streamline development. Maven handles backend build processes, dependency management, and project configuration. Together, they ensure efficient and organized development workflows.





# MODULE DESCRIPTION

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**Module 1: Authentication & User Management :** Handles role-based registration for Students, Employers, and Admins with secure login using JWT-based authentication via Spring Security. It also supports profile management and role-specific access control to ensure data security and personalized user experience.

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**Module 2: Job & Application Processing :** Implements the core recruitment workflow, enabling employers to create, update, and manage job listings efficiently. Students can apply for jobs, upload resumes, and track application status in real time through an integrated system.

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**Module 3: Database & File Management :** Manages structured storage of users, jobs, and application data using MySQL for consistency and reliability. It also ensures secure upload, storage, and retrieval of resume files with proper data handling mechanisms.

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**Module 4: Innovative Integrations (AI & Alerts) :** Integrates the Google Gemini API to generate dynamic, role-specific interview questions for employers. Uses SMTP-based email services for automated notifications, alerts, and OTP verification to enhance communication.

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**Module 5: Dashboards & Reporting :** Provides role-based dashboards with a modern Neo-Brutalist UI for students, employers, and admins. It offers real-time updates, candidate shortlisting views, and data export features (CSV) for better analysis and decision-making.

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# IMPLEMENTATION SCREENSHOTS

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**Elevate**

Login Register

Email Address

you@example.com

Password

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Login to Account



# IMPLEMENTATION SCREENSHOTS

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8(A)

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The screenshot displays the Elevate job portal dashboard. The top navigation bar includes the Elevate logo, a bell icon, and links for Dashboard, Find Jobs, Logout, and Post a Job. The main content area features five job listings arranged in a grid:

- senior designer** by **infosys**: Location: chennai, Salary: \$120k. Job description: job. Button: Apply Now.
- infosys technova**: Location: chennai, Salary: \$120k. Job description: good job. Button: Apply Now.
- Backend Developer** by **DataSync Inc**: Location: New York, NY, Salary: \$130k - \$160k. Job description: Looking for a strong Node.js developer to build scalable microservices and robust RESTful APIs. Button: Apply Now.
- Product Designer** by **Creative Studio X**: Location: Remote, Salary: \$110k - \$130k. Job description: Join our creative team to design beautiful, intuitive user experiences for our next-generation web applications. Button: Apply Now.
- Senior Frontend Engineer** by **TechNova Solutions**: Location: San Francisco, CA (Hybrid), Salary: \$140k - \$170k. Job description: We are looking for a Senior Frontend Engineer with deep expertise in React and modern CSS to lead our core UI team. Button: Apply Now.



# IMPLEMENTATION SCREENSHOTS

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- Admin Login Page

The screenshot shows the Admin Login Page for the Elevate application. The page has a dark blue background. At the top, there is a logo with a bell icon and the word "Elevate". Below the logo, there are two buttons: "Login" (highlighted in blue) and "Register". Underneath these buttons, there are two input fields: "Email Address" with the text "admin@elevate.com" and "Password" with masked characters ".....". At the bottom, there is a large blue button labeled "Login to Account".

- User Login Page

The screenshot shows the User Login Page for the Elevate application. The page has a dark blue background. At the top, there is a logo with a bell icon and the word "Elevate". Below the logo, there are two buttons: "Login" (highlighted in blue) and "Register". Underneath these buttons, there are two input fields: "Email Address" with the text "you@example.com" and "Password" with masked characters ".....". At the bottom, there is a large blue button labeled "Login to Account".

# IMPLEMENTATION SCREENSHOTS



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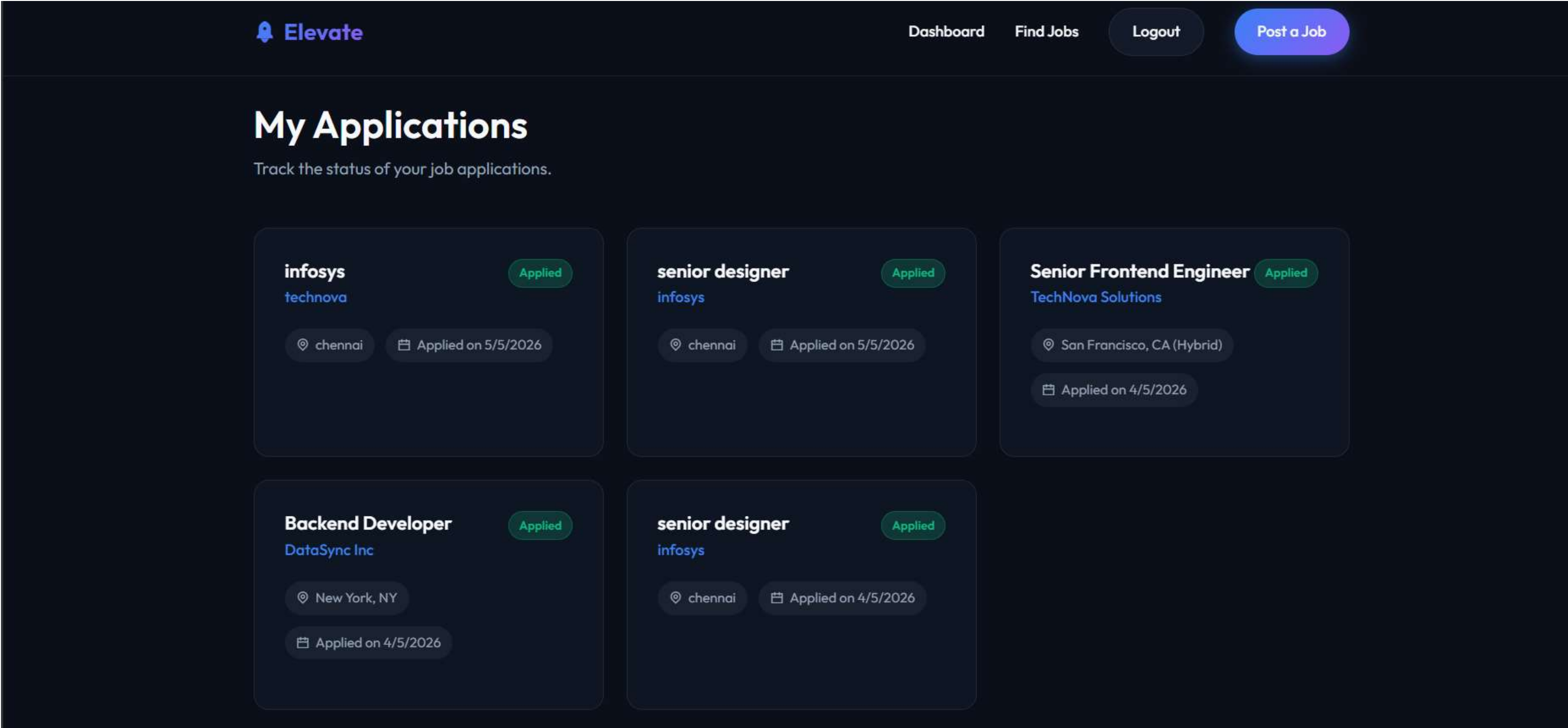
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# IMPLEMENTATION SCREENSHOTS



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Dashboard

Find Jobs

Logout

Post a Job

## Hiring? Post a Job

Reach thousands of professionals actively looking for their next role.

Job Title

e.g. Senior Product Designer

Company Name

e.g. TechNova

Location

e.g. San Francisco (Remote)

Salary Range (Optional)

e.g. \$120k - \$150k

Job Description

Describe the role and responsibilities...

Publish Listing →

# IMPLEMENTATION SCREENSHOTS



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The screenshot displays the Gmail interface. On the left, the navigation sidebar includes 'Mail' (99+), 'Chat' (1), and 'Meet'. The main inbox area shows a list of folders: 'Compose', 'Inbox' (2,710), 'Starred', 'Snoozed', 'Sent', 'Drafts', 'Categories', and 'More'. The email content area shows two messages. The first message is from 'vt26363@veltech.edu.in' to 'pallavi', with a subject 'Application Received: infosys at technova'. The body text reads: 'Hi pallavi, Thank you for applying for the infosys position at technova! We have received your application and resume. Our team will review your qualifications and get back to you soon. Best regards, The Job Portal Team'. The second message is a 'Mail Delivery Subsystem' notification stating 'Recipient inbox full' and 'Your message couldn't be delivered to pallavi@gmail.com. Their inbox is full or it's getting too much mail right now.' The bottom of the email shows action buttons: 'Reply', 'Forward', 'Share in chat', and 'Share in chat'.



# IMPLEMENTATION SCREENSHOTS

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```
38 • SELECT * FROM applications;
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```

|   | id   | job_id | name | email                 | resume_link                        | timestamp           |
|---|------|--------|------|-----------------------|------------------------------------|---------------------|
| ▶ | 1    | 3      | sai  | aakashsai35@gmail.com | https://linkedin.com/in/sai-aakash | 2026-05-03 20:27:19 |
| • | NULL | NULL   | NULL | NULL                  | NULL                               | NULL                |

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

Result Grid

Form Editor

Field Types

# ADVANTAGES & LIMITATIONS



1 **Advantages Unified Workflow** : Centralizes job posting, applications, and resume management in one platform.

2 **AI Productivity** : Uses Google Gemini API to generate role-specific interview questions instantly.

3 **Modern UI/UX** : Neo-Brutalist design with role-based dashboards for a clean, engaging experience.

4 **Automated Communication** : SMTP-based emails provide instant updates and OTP verification.

5 **Strong Security** : Spring Security with JWT ensures secure authentication and role-based access.

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## Limitations

8 **API Dependency** : Relies on external services like Gemini API and SMTP, which may fail during downtime or limits.

9 **AI Accuracy** : Generated questions may sometimes be complex or slightly inaccurate, requiring manual review.

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***Thank you***